

Heart Disease Prediction and Detection Using AI Techniques

Client Technical Report - experimental progress, added work, hybrid models, and results.

Executive Summary

The project was extended from a basic single-dataset experiment into a thesis-ready comparative framework. We evaluated Kaggle CVD, UCI Heart Disease, and OpenML Heart Disease Comprehensive 1190, then built tuned boosting models and hybrid ensemble models.

Best repeated CV accuracy: Tuned CatBoostClassifier at 94.26% accuracy, 94.56% F1-score, and 96.78% ROC-AUC.

Best repeated CV ROC-AUC: Tuned Soft Voting Hybrid at 97.21% ROC-AUC and 96.63% PR-AUC.

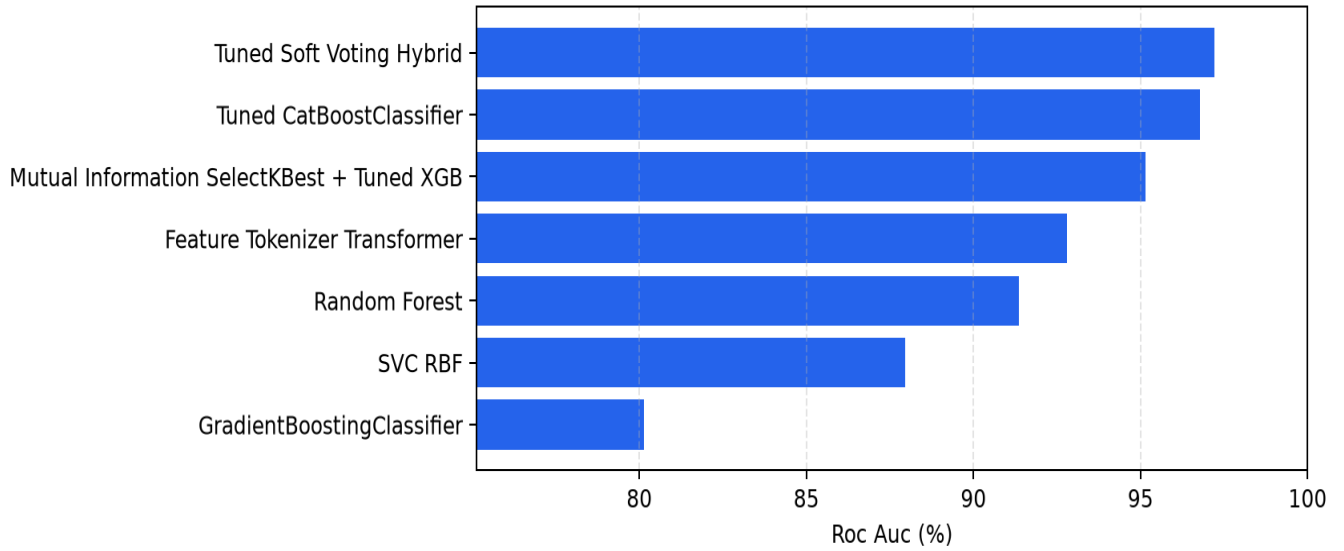
Work Completed

- Reviewed the proposal topic and translated it into an executable ML plan.
- Loaded and cleaned the original Kaggle cardiovascular dataset.
- Benchmarked classical ML, boosting, and neural baseline models.
- Added UCI and OpenML heart disease benchmarks for stronger thesis evidence.
- Built tuned CatBoost, LightGBM, XGBoost, Random Forest, and Extra Trees models.
- Built Soft Voting and Stacking hybrid ensemble models.
- Added SHAP explainability, feature selection analysis, and FT-Transformer.
- Created one clean notebook where metrics are produced by execution.

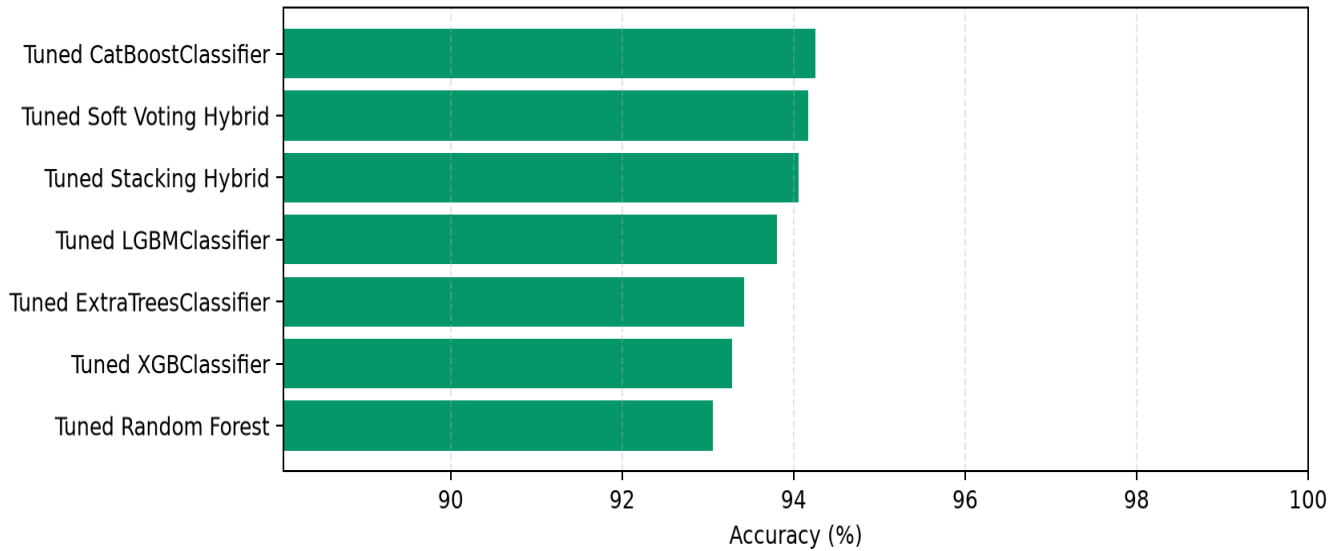
Dataset Comparison

dataset	model	accuracy	f1	recall	roc_auc
OpenML Heart 1190	Tuned Soft Voting Hybrid	94.17%	94.52%	95.23%	97.21%
OpenML Heart 1190	Tuned CatBoostClassifier	94.26%	94.56%	94.86%	96.78%
OpenML Heart 1190 + Feature Selection	Mutual Information SelectKBest + Tuned XGB	90.56%	91.00%	90.24%	95.14%
OpenML Heart 1190 + FT-Transformer	Feature Tokenizer Transformer	88.24%	89.23%	92.06%	92.80%
UCI Cleveland	Random Forest	83.49%	81.42%	79.10%	91.37%
UCI All Processed	SVC RBF	82.03%	83.64%	83.07%	87.95%
Kaggle CVD 70k	GradientBoostingClassifier	72.23%	73.93%	79.58%	80.13%

Dataset and Model ROC-AUC Comparison



Tuned Model Accuracy Comparison



OpenML Baseline Models

model	accuracy	accuracy_std	precision	recall	f1	roc_auc	roc_auc_std	pr_auc
XGBClassifier	92.02%	1.85%	93.05%	91.89%	92.40%	96.28%	1.06%	95.58%
CatBoostClassifier	91.76%	1.95%	92.28%	92.21%	92.20%	96.40%	1.13%	95.41%
Random Forest	91.09%	1.96%	91.47%	91.89%	91.60%	96.36%	1.11%	95.81%
ExtraTreesClassifier	88.99%	2.36%	89.24%	90.14%	89.64%	95.49%	1.14%	95.49%
LGBMClassifier	88.99%	2.30%	89.88%	89.35%	89.55%	94.96%	1.36%	94.65%
SVC RBF	87.23%	2.31%	87.28%	88.87%	88.03%	93.80%	1.64%	93.70%
GradientBoostingClassifier	86.39%	2.91%	86.41%	88.23%	87.26%	93.37%	1.37%	93.28%
Logistic Regression	84.71%	2.65%	85.23%	86.16%	85.63%	91.73%	2.05%	92.43%

Tuned Models and Hybrid Ensembles

model	accuracy	accuracy_std	precision	recall	f1	roc_auc	roc_auc_std	pr_auc
Tuned CatBoostClassifier	94.26%	2.08%	94.38%	94.86%	94.56%	96.78%	1.55%	95.86%
Tuned Soft Voting Hybrid	94.17%	1.59%	93.89%	95.23%	94.52%	97.21%	1.43%	96.63%
Tuned Stacking Hybrid	94.06%	1.68%	94.03%	94.86%	94.40%	97.19%	1.44%	96.61%
Tuned LGBMClassifier	93.81%	1.72%	93.90%	94.49%	94.15%	96.90%	1.60%	96.00%
Tuned ExtraTreesClassifier	93.42%	1.55%	93.13%	94.60%	93.81%	96.92%	1.05%	96.50%
Tuned XGBClassifier	93.28%	1.76%	93.28%	94.12%	93.66%	96.37%	1.63%	95.41%
Tuned Random Forest	93.05%	1.83%	92.96%	94.06%	93.47%	96.75%	1.45%	96.08%

Feature Selection Experiments

model	accuracy	accuracy_std	precision	recall	f1	roc_auc	roc_auc_std	pr_auc
Mutual Information SelectKBest + Tuned XGB	90.56%	2.40%	91.87%	90.24%	91.00%	95.14%	2.02%	94.22%
ANOVA SelectKBest + Tuned CatBoost	88.49%	2.76%	89.71%	88.49%	89.00%	93.74%	2.09%	93.38%
ANOVA SelectKBest + Tuned XGB	87.42%	2.96%	89.12%	86.96%	87.94%	92.71%	2.61%	92.72%

Feature Tokenizer Transformer

model	epochs	best_valid_auc	accuracy	precision	recall	f1	roc_auc	pr_auc
Feature Tokenizer Transformer	24	94.07%	88.24%	86.57%	92.06%	89.23%	92.80%	90.33%

Final Holdout Evaluation

model	accuracy	precision	recall	f1	roc_auc	pr_auc
Tuned Stacking Hybrid	92.86%	95.04%	91.27%	93.12%	97.40%	96.90%
Tuned Soft Voting Hybrid	92.86%	94.31%	92.06%	93.17%	97.36%	96.91%
Tuned CatBoostClassifier	92.44%	95.00%	90.48%	92.68%	96.75%	96.08%

Hybrid Model Design

The Soft Voting Hybrid combines calibrated probability outputs from multiple strong learners. The Stacking Hybrid trains a meta-model on base model predictions. These designs were selected because boosting models and tree ensembles showed complementary strengths across accuracy, F1-score, ROC-AUC, and PR-AUC.

Comparison With the Published Baseline

	method	accuracy	sensitivity_recall	roc_auc	comparison_type
	Stacking ensemble	91%	Not reported in summary line	0.92	Same family of heart-disease benchmark; protocol differs
	Stacking ensemble	98%	Not reported in summary line	0.97	Same family of heart-disease benchmark; protocol differs
	Bayesian-optimized XGBoost + SHAP	98.40%	98.90%	Not reported in Table 5	Dataset overlaps with UCI Cleveland; protocol differs
	Bayesian-optimized XGBoost + SHAP	95.90%	97.50%	Not reported in Table 6	Different dataset; indirect comparison only
	Tuned CatBoostClassifier	94.26%	94.86%	0.9678	Our reproducible repeated-CV result
	Tuned Soft Voting Hybrid	94.17%	95.23%	0.9721	Our reproducible repeated-CV hybrid result
FT-Transformer	Feature Tokenizer Transformer	88.24%	92.06%	0.9280	Our deep tabular baseline

Direct comparison is valid only when the same dataset, split strategy, preprocessing, and evaluation protocol are used. Published values are therefore used for literature positioning, while notebook-generated metrics are reported as reproducible results.

Research Contribution Positioning

	project_response	evidence
is not a sufficient contribution.	Treat multiple datasets as validation evidence, not as the main novelty.	Kaggle, UCI, OpenML, feature-selection, and FT-Transformer results are
is required.	Propose a unified explainable hybrid framework.	Soft Voting and Stacking hybrids combine tuned boosting and tree-enser
ack box.	Add SHAP explainability for the final tree-based model.	SHAP identifies chest pain type, ST slope, oldpeak, cholesterol, max hea
esented if the thesis claims modern AI.	Add Feature Tokenizer Transformer for tabular clinical data.	FT-Transformer is evaluated as a deep tabular baseline and compared w
should be evaluated, not assumed.	Add ANOVA and mutual-information SelectKBest experiments.	Feature selection is shown to reduce dimensionality but does not outper
y come from different protocols.	Separate published results from reproducible notebook-generated results.	Published-work comparison table includes a comparison-type note for ea

The main contribution is the unified explainable hybrid framework, not the use of multiple datasets alone. Multiple datasets are used as validation evidence for generalization.

SHAP Explainability

feature	mean_abs_shap
chest_pain_type_4.0	1.0901115744116048
ST_slope_2.0	0.9824976081888284
oldpeak	0.8601912077693442
ST_slope_1.0	0.8558491137864163
cholesterol	0.7732620850698556
max_heart_rate	0.5968171613736415
exercise_angina_1.0	0.5889404130214758
sex_1.0	0.5165384956746343
age	0.494082307021947
resting_bp_s	0.4269255127930438

feature	mean_abs_shap
sex_0.0	0.354253630880887
resting_ecg_2.0	0.3233158089041967

SHAP was added to explain the clinical factors driving the final tree-based model. The strongest drivers included chest pain type, ST slope, oldpeak, cholesterol, maximum heart rate, exercise angina, sex, age, and resting blood pressure.

Recommended Next Steps

- Add visible hyperparameter search inside the notebook if required.
- Optionally add nested CV for the optimized XGBoost comparison.
- Add a methodology diagram for thesis writing.
- Export the approved notebook/report to final PDF.